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14 August 1967
SUPERSEDING
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4 August 1967

MILITARY SPECIFICATION

PROJECTILE ASSEMBLY FOR CARTRIDGE, H.E., 81MM, M374

1. SCOPE

1.1 This specification covers Projectile Assembly for Cartridge, H.E., 81MM, M374.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Military

MIL-A-2550	- Ammunition and Special Weapons, General Specification for
MIL-M-11472	- Magnetic-Particle Inspection, Processes for Ferromagnetic Materials

STANDARDS

Federal

Federal Test Method Standard No. 151 - Metals, Test Methods

Military

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	- Quality Assurance Terms and Definitions
MIL-STD-410	- Qualification of Inspection Personnel (Magnetic Particle and Penetrant)
MIL-STD-1167	- Ammunition Data Card
MIL-STD-1169	- Packaging, Packing and Marking for Shipment of Inert Ammunition Components

FSC 1315

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2.1 STANDARDS (Cont'd)

Military

MIL-STD-1235	- Single and Multilevel Continuous Sampling Procedure and Tables for Inspection by Attributes

DRAWINGS

Munitions Command

C10520203	- Projectile, 81MM, H.E., M374 Assembly
IEL-10520203	- Inspection Equipment List for Projectile, 81MM, H.E., M374 Assembly

PUBLICATIONS

PA-PD-792	- Equipment, Testing or Measuring for Ammunition Items

Frankford Arsenal

10519685	- Operating Instructions Manual for the Ultrasonic Inspection of Body Castings for 81MM Projectile
ASA-B46.1	- Surface Texture, Surface Roughness, Waviness and Lay

(Copies of specifications, standards, drawings and publications required by supplier in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General.- Materials, parts and assemblies shall comply with the requirements specified on the applicable drawings, referenced specifications and the following:

3.2 Pilot heat treatment sample (Body and Body Alternative I).- Prior to the submittal of the initial production sample, a pilot heat treatment sample shall be required to qualify heat treatment procedure. After qualification of the heat treatment procedure no change in procedure will be permitted without prior approval of the cognizant technical agency. A copy of the complete heat treatment procedure shall accompany the pilot heat treatment sample.

3.3 Heat treatment.

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3.3.1 Forging or cold shaping (Body and Body Alternative I).-Each projectile body shall be heat treated by heating to a specific temperature between the upper and lower critical transformation temperature (1330 degrees F to 1430 degrees F). The specific temperature must be selected to assure meeting the minimum mechanical properties required on the drawing.

3.3.2 Casting.-The heat treatment of the body casting shall be uniform throughout each lot and shall be so conducted as to produce in each body casting machanical properties specified on the applicable drawing.

3.3.3 Forging (Body Alternatives III and IV).-Each projectile body shall be heat treated above the upper critical temperature, not to exceed 1600 degrees F., oil quenched and tempered uniformly to meet the mechanical properties specified on the applicable drawing. No work involving plastic flow is permitted after tempering unless followed by a stress relief.

3.3.4 Cold shaping (Body Alternatives III and IV).-Each cold worked body shall be stress relieved at a minimum temperature of 700 degrees F. and held at temperature for a minimum of 30 minutes. No work involving plastic flow of metal is permitted after final stress relief.

3.4 Mechanical properties.-All components shall meet the mechanical properties requirements as specified on the applicable drawings.

3.5 Metal defects.

3.5.1 Forging or cold shaping.-All components shall be free of cracks, seams, cold shuts, imbedded scale, porosity, inclusions and other similar defects.

3.5.2 Casting.-The machined body casting shall be free from cracks, cold shuts, hot tears, shrinkage, inclusions, blowholes, porosity or other defects.

3.5.3 Base adapter.-The base adapter shall be free from cracks, splits, cold shuts, laminations, porosity or similar defects.

3.6 Brazed joint (Alternative Body Assemblies).-The braze metal shall be visible around the entire area where the base adapter joins the body wall. The braze metal shall also be visible for the 360 degrees around the joint between the rear of the body and the base adapter flange.

3.7 Hydrostatic test (Alternative Body Assemblies).-The body assembly shall withstand the stresses developed by the application of 5000 pounds per square inch (psi) minimum, internal hydrostatic pressure for five seconds without leakage, rupture, or permanent deformation.

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3.8 Air pressure test (Alternative Body Assemblies).- After hydrostatic testing, the body assembly shall withstand an internal air pressure of 150 psi minimum for 15 seconds minimum without evidence of leakage.

3.9 Joint quality (Alternative Body Assemblies).- The brazed joint shall be free of entrapped slag, voids, unwetted or unbonded areas and other brazing defects. The aggregate area of these defects shall not exceed 25 percent of the area of the faying surfaces; and the maximum extent of a defect shall not exceed 50 percent of the overlap distance (width) of the joint.

3.10 Protective coating.- Protective coating shall be in accordance with the applicable drawing and the specifications referenced thereon.

3.11 Proving Ground functional tests.- The projectile body assembly shall reveal no evidence of failure when tested at excess and service pressure. Body assembly failure shall be:

a. Projectile body and base adapter deformation attributable to noncompliance with drawing and specification requirements.

b. Loss or breakup of any component, except the obturator in the mortar tube or in flight.

3.12 Workmanship.- The requirements for workmanship are as shown on the applicable drawings in the applicable specifications and the following:

3.12.1 Surface finish.- The requirements for surface finish are as detailed on the applicable drawings. The cavity shall be free of burrs, scale, fins, deep seams, pits or other surface discontinuities.

3.12.2 Threads.- Threads shall be full and undamaged for the entire minimum length as required by the applicable drawings and shall be in accordance with the provisions of MIL-A-2550.

3.12.3 Burr.- No part shall have a burr which might interfere with the assembly or function of the round or which might be injurious to personnel handling the item.

3.12.4 Foreign matter.- No part or assembly shall contain dirt, grease, chips, rust, corrosion, or other foreign matter. Particular attention shall be given to the cavity of the body assembly, to insure that no grit, scale, accumulated braze alloy, residual flux or other foreign matter remains therein after final cleaning.

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3.13 Weight control.- Projectile assembly (prior to paint and insertion of cap) shall be segregated into two weight groups:

a. 4.880 to 5.180 pounds

b. 5.181 to 5.280 pounds

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Quality Assurance terms and definitions.- Reference shall be made to MIL-STD-109 to define quality assurance terms used.

4.1.2 Inspection.- Inspection shall be in accordance with MIL-A-2550, except as specified herein.

4.1.3 Inspection criteria.

a. Successful passage of all inspections required by the drawings and specifications shall be considered as a condition precedent to performing the Proving Ground functional tests (see 3.11)

b. Acceptance or rejection of a lot by the Government shall be contingent on successful passage of the Proving Ground functional tests.

4.2 First article sample.

4.2.1 Initial production sample.- At the beginning of regular production, a sample shall be submitted in accordance with contract requirements and shall consist of 70 metal parts assemblies as follows:

55 projectile assemblies complete, painted
5 projectile assemblies complete, unpainted
10 sets of component parts, unassembled

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4.2.1 (Cont'd)

Fifty of the above-painted projectile assemblies shall be forwarded to a designated proving ground for ballistic testing. The sample shall be manufactured using the same materials, equipment, processes and procedures as used in regular production. All parts and materials, including packaging and packing, shall be the same as used for the regular production and shall be obtained from the same source of supply.

4.2.1.1 Examination and test.- After inspection and provisional acceptance at source, the sample shall be inspected for all requirements of the drawings and specifications at a Government laboratory or such other facility specified in the contract.

4.2.1.2 Initial production sample failure.- Failure of the sample to comply with requirements of the drawings and specifications shall result in sample disapproval.

4.2.2 Pilot heat treatment sample (Body and Body Alternative I).- This sample shall consist of 10 heat treated projectile bodies selected from a production furnace charge when a batch type furnace is used or 10 heat treated projectile bodies selected from a continuous furnace. In addition, 10 projectile bodies that have not been heat treated will be submitted.

4.3 Inspection.

4.3.1 Lot.

4.3.1.1 Submission of product.- The product shall be submitted in accordance with MIL-STD-105 or MIL-STD-1235, as applicable.

4.3.1.2 Heat treatment lots.

4.3.1.2.1 Forging or cold shaping (Body and Body Alternative I).- Forging or cold shaping heat treatment lots shall contain bodies, from not more than one heat of steel, that have been treated in the same heat treating equipment at the same temperature in one unchanged process. The forging or cold shaping lot thus constituted shall form the basis for selecting the sample body forgings or cold shapings for the tension test hereinafter prescribed.

4.3.1.2.2 Castings.

4.3.1.2.2.1 Pearlitic malleable iron.- For identification purposes, any quantity of body castings (but not to exceed 4-hour production of the final heat treatment furnace that has been subjected to final heat treatment in the same furnace, at the same temperature, for the same length of time or conveyor speed) shall be considered as a casting lot. The casting lot thus constituted shall form the basis for selecting sample body castings for the tension test hereinafter prescribed.

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4.3.1.2.2.2 Ductile iron.- A heat treatment lot shall consist of one or more casting lots not to exceed four hours of continuous foundry production or a maximum of 5,000 items whichever is greater. All castings in a heat treatment lot shall be subjected to the same heat treatment cycle (temperature and time) in the same type of furnace equipment. The lot thus constituted shall be the basis for selection of sample castings to be used for the tension tests.

4.3.1.2.3 Forging (Body Alternatives III and IV).- Forging lots shall contain body forgings from not more than one heat of steel that have been heat treated in the same heat treating equipment at the same temperature in one unchanged process. The forging lot thus constituted shall form the basis for selecting sample body forgings for the tension test hereinafter prescribed.

4.3.1.2.4 Cold shaping (Body Alternatives III and IV).- Cold shaping lots shall contain cold worked bodies from not more than one heat of steel that have been cold worked by the same process utilizing the same tooling and equipment and stress relieved in the same furnace at the same temperature for the same length of time. The cold shaping lot thus constituted shall be the basis for the selection of samples for the tension test hereinafter prescribed.

4.3.1.2.5 Miscellaneous.- Small lots, such as bodies from abnormally small heats of steel or bodies which have become separated from previously delivered lots, etc., may, at the discretion of the manufacturer, be grouped into miscellaneous forging, cold shaping or casting lots as the case may be. Such lots shall be identified by a miscellaneous lot code number.

4.3.1.6 Grouping.- Forging, cold shaping or casting lots shall remain grouped together for lot identity purposes, according to forging, cold shaping or casting lots, until such time as the results of the tension test have indicated that the lot has complied with the mechanical properties requirements of the applicable drawing.

4.3.1.7 Projectile lot.- All forging, casting or cold shaping lots that have been determined to have complied with the mechanical properties requirements of the applicable component drawing, may be re-grouped for assembly into projectile lots. A projectile lot shall consist of projectile assemblies produced by one manufacturer, under one contract, in one unchanged process, in accordance with the same drawing, same drawing revision, same specification and same specification revision. A projectile lot shall also consist of one weight group as specified in 3.13.

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4.3.2 Examination.- One hundred percent examination shall be performed for all critical defects. Examination for major and minor defects shall be performed on a defect basis in accordance with the classification of defects contained herein, using the sampling plans contained in MIL-STD-105. Continuous sampling plans in accordance with MIL-STD-1235 may be used if approved by the procuring activity. An AQL of 0.40 percent for each major defect and an AQL of 0.65 percent for each minor defect shall be used. All nonconforming material shall be rejected. Four power (4X) magnification may be used in performing visual examination. The roughness comparison specimens prescribed by ASA-B46.1 shall be used as a basis for surface roughness determination.

4.3.2.1 Adapter, Base, Drawing C10535863 covering a detail of Drawing C10520203

	Defect	Method of Inspection
CRITICAL		
	None defined	
MAJOR		
101.	Diameter of brazing surface	Gage
102.	Diameter of shoulder	Gage
103.	Thickness through base	Gage
MINOR		
201.	Thickness of shoulder (min)	Gage
202.	Length, total	Gage
203.	Angle of shoulder (when applicable)	Gage
204.	Radii or chamfers missing or incorrect	Visual (See 6.1)
205.	Surface finish improper	Visual (See 6.1)

4.3.2.2 Body Alternative I Drawing F10535871, Body Alternative II Drawing D10535873, Body Alternative IV Drawing F10535893 and Body Alternative V Drawing F10535961 covering a detail of Drawing C10520203

CRITICAL

None defined

MAJOR

None defined

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4.3.2.2 (Cont'd)

Defect	Method of Inspection
MINOR	
201. Length, total	Gage
202. Diameter at rear of body	Gage
203. Radius at rear of body missing or incorrect	Visual

4.3.2.3 Body, Drawing F10535869 and Body Alternative III Drawing F10535892 covering a detail of Drawing C10520203

CRITICAL

None defined

MAJOR

101. Thickness through base	Gage

MINOR

None defined

4.3.2.4 Projectile subassembly (prior to paint and insertion of cap), Drawing D10520202 covering a detail of Drawing C10520203

CRITICAL

1. Braze metal not visible for full 360 degrees at juncture of base adapter and body wall (alternative body assembly only)	Visual
2. Braze metal not visible for full 360 degrees at joint between the rear face of body and the adapter flange (alternative body assembly only)	Visual
3. Hydrostatic or air pressure test stamps missing (indicating failure of the assembly to comply with one or more of the requirements) (alternative body assembly only)	Visual
4. Ultrasonic or magnetic particle stamps missing (indicating failure of the body to comply with one or more of the requirements) (Alternative body design No. II only)	Visual
5. Metal defective	Visual

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4.3.2.4 (Cont'd)

	Defect	Method of Inspection
MAJOR		
101.	Weight	Approved Scale
102.	Concentricity of pitch diameter of nose thread with bourrelet, including perpendicularity of front face	Gage
103.	Concentricity of band groove with bourrelet	Gage
104.	Concentricity at rear of body with bourrelet	Gage
105.	Concentricity of ogive with bourrelet	Gage
106.	Concentricity of pitch diameter of base thread with bourrelet, including perpendicularity of rear face	Gage
107.	Diameter of band groove	Gage
108.	Diameter of bourrelet, min.	Gage
109.	Thickness of wall at bourrelet	Gage
110.	Thickness of wall between bourrelet and taper, min.	Gage
111.	Thickness of wall at front of taper	Gage
112.	Thickness of wall at rear of taper	Gage
113.	Thickness through base at juncture of thread boss and rear face, min.	Gage
114.	Width of band groove	Gage
115.	Variation in wall thickness	Gage
116.	Base cover weld or braze, inadequate	Manual
117.	Surface finish of cavity, improper	Visual (see 6.1)
118.	Foreign matter in cavity	Visual
MINOR		
201.	Diameter of nose, outside	Gage
202.	Diameter at ogive basic length	Gage
203.	Diameter at rear bourrelet basic length	Gage
204.	Diameter at basic length on taper	Gage
205.	Diameter at rear basic length on taper (one piece design)	Gage
206.	Diameter of base thread undercut	Gage
207.	Diameter of base cover	Gage
208.	Diameter of base cover, min. (if applicable)	Gage
209.	Length of thread boss (including base cover)	Gage
210.	Length from front face to rear of shoulder (one piece designs)	Gage
211.	Length of nose thread cavity	Gage
212.	Length, total, max.	Gage

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4.3.2.4 (Cont'd)

Defect	Method of Inspection
MINOR (Cont'd)	
213. Location of band groove	Gage
214. Radii or chamfers missing or incorrect	Visual (see 6.1)
215. Surface finish improper (except as otherwise specified)	Visual (see 6.1)
216. Overlap between base adapter and body (when applicable)	Visual
217. Burr	Visual
218. Foreign matter (except as otherwise specified	Visual

4.3.2.5 Projectile subassembly (after paint) Drawing D10520202 covering a detail of Drawing C10520203

CRITICAL	
None defined	
MAJOR	
101. Pitch diameter of nose thread (max)	Gage
102. Minor diameter of nose thread (max)	Gage
103. Pitch diameter of base thread (min)	Gage
104. Major diameter of base thread (min)	Gage
105. Diameter of bourrelet, max.	Gage
106. Length of effective nose thread, min.	Gage
107. Length of effective base thread, min.	Gage
108. Paint in cavity inadequate	Visual
109. Pool of paint in cavity	Visual
110. Thread(s) damaged or not full	Visual
111. Foreign matter	Visual
112. Protector cap missing	Visual
MINOR	
201. Pitch diameter of nose thread (min)	Gage
202. Minor diameter of nose thread (min)	Gage
203. Pitch diameter of base thread (max)	Gage
204. Major diameter of base thread (max)	Gage
205. Corrosion preventive compound missing from threads	Visual
206. Paint on thread(s)	Visual
207. Paint not dry	Visual/Tactile
208. Paint inadequate or damaged	Visual
209. Marking missing, incorrect or illegible	Visual

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4.3.2.6 Band, Obturating, Drawing C10520201 covering a detail of Drawing C10520203

Defect	Method of Inspection
CRITICAL	
None defined	
MAJOR	
101. Diameter, inside	Gage
102. Width	Gage
MINOR	
201. Grooves, missing or incorrect	Visual (see 6.1)
202. Material defective	Visual
203. Flash	Visual

4.3.2.7 Projectile assembly, Drawing C10520203

CRITICAL	
None defined	
MAJOR	
101. Concentricity of obturator band with bourrelet	Gage
102. Diameter over obturator band	Gage
103. Obturator band damaged	Visual
MINOR	
None defined	

4.3.2.8 Packed carton (prior to sealing)

CRITICAL	
None defined	
MAJOR	
None defined	

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4.3.2.8 (Cont'd)

Defect	Method of Inspection
MINOR	
201. Support missing or damaged	Visual
202. Separator missing or damaged	Visual
203. Improper assembly	Visual
204. Incorrect number in carton	Visual

4.3.2.9 Packed carton (after sealing)

CRITICAL

None defined

MAJOR

None defined

MINOR	
201. Carton damaged, cut or gouged through all layers	Visual
202. Carton not completely sealed	Visual
203. Contents loose	Visual/Manual
204. Marking missing, incorrect or illegible	Visual

4.3.3 Inspection testing.- The following tests shall be performed in accordance with the provisions of 4.4.

4.3.3.1 Heat treatment qualification.

4.3.3.1.1 Forgings or cold shapings (Body and Body Alternative I).- The sample furnished in accordance with 4.2.2 shall be examined by the cognizant technical agency to determine compliance with 3.3.1. Failure to comply shall be cause for rejection and resubmission of additional samples will be required until such time as an acceptable heat treatment procedure is obtained.

4.3.3.2 Mechanical properties.

4.3.3.2.1 Hardness.

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4.3.3.2.1.1 Forging or cold shapings (Body and Body Alternative I).- In order to determine compliance with the hardness requirement of the applicable drawing, each heat treated body will be subjected to a hardness test with a Rockwell Hardness Tester. Failure to comply shall be cause for rejection of the body.

4.3.3.2.1.2 Castings.-In order to determine uniformity of heat treatment, each heat treated body will be subjected to a hardness test with a Brinell Hardness Tester using a 5 second minimum dwell or by any other method approved by the procuring activity.

4.3.3.2.1.3 Forgings (Body Alternatives III and IV).-In order to determine uniformity of heat treatment, each body forging in the forging heat treatment lot (as defined by 4.3.1.2.3) shall be subjected to a hardness test with a Brinell Hardness Tester or by any other method of hardness testing approved by the procuring activity.

4.3.3.2.1.4 Cold shapings (Body Alternatives III and IV).- Twenty-five cold shaped bodies from the first batch of each cold shaping lot (as defined by 4.3.1.2.4) processed through the stress relief furnace and five bodies from each hour's production thereafter for the balance of the lot shall be subjected to a hardness test with a Rockwell Hardness Tester or by any other method approved by the procuring activity, to assess uniformity of the product. Hardness values shall be recorded.

4.3.3.2.1.5 Miscellaneous.-Each miscellaneous lot shall be subjected to the hardness tests listed in paragraphs 4.3.3.2.1.1, 4.3.3.2.1.2, 4.3.3.2.1.3 or 4.3.3.2.1.4 as applicable.

4.3.3.2.2 Tension test.

4.3.3.2.2.1 Forging or cold shaping (Body and Body Alternative I).- Two projectile bodies, one of the hardest and one of the softest from each forging or cold shaping heat treatment lot (as defined in paragraphs 4.3.1.2.1 or 4.3.1.2.5 as applicable) shall be subjected to this test to determine compliance with the mechanical properties required on the applicable drawing. Failure of any tension test specimen to comply with the requirements shall be cause for rejection of the lot represented by the specimens.

4.3.3.2.2.2 Casting.-Two body castings, one of the hardest and one of the softest as determined by the hardness test shall be subjected to this test. Failure of any tension test specimens to comply with the mechanical property requirement of the applicable drawing shall be cause for rejection of the lot.

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4.3.3.2.2.3 Forging (Body Alternatives III and IV).-Two body forgings, one of the hardest and one of the softest as determined by the hardness test shall be subjected to this test. Failure of any tension test specimen to comply with the mechanical property requirements of the applicable drawing shall be cause for rejection of the lot. At the option of the supplier a production control sample may be utilized to establish the hardness range. When a production control sample is used the hardness range thus established shall be maintained throughout the lot. The size of the sample shall be determined by the supplier. Projectile bodies which fall outside of the established range may be re-heat treated to fall within the range or new mechanical properties may be determined from newly selected projectile bodies which will represent the true extremes of hardness in the heat treatment lot.

4.3.3.2.2.4 Cold shaping (Body Alternatives III and IV).-Two projectile bodies from each cold shaping heat treatment lot shall be selected at random and subjected to this test. Failure of any tension test specimens to comply with the mechanical property requirements of the applicable drawing shall be cause for rejection of the lot.

4.3.3.2.2.5 Miscellaneous lots.-When miscellaneous lots are used, double the sample size will be required.

4.3.3.3 Metal defects.-

(4.3.3.3.1 through 4.3.3.3.2.1.2 are critical defects and apply to Body Alternative II and Body Alternative V).

4.3.3.3.1 Ultrasonic inspection.-Each body casting, after rough machining of the outside surface, shall be subjected to an automatic ultrasonic inspection in accordance with Manual 10519685. Rejection of any body casting, by the ultrasonic testing machine shall be considered as failure to comply with the requirements of 3.5.2 and shall be cause for rejection of that body casting. To identify body castings that have passed this test, the letter "U" shall be indented thereon by a metal stamp.

4.3.3.3.2 Magnetic particle inspection.

4.3.3.3.2.1 Upon initiation of production, the contractor shall perform 100 percent magnetic particle inspection on all body castings that have passed the ultrasonic inspection. The magnetic particle inspection shall be performed in accordance with the contractor's approved precedure and shall proceed through each cycle: i.e., magnetize, apply suspension, drain, visually inspect, demagnetize and clean.

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4.3.3.3.2.1.1 If at any time a magnetic particle indication is present and the contractor is in doubt as to whether is is a true indication or not, he will be permitted to demagnetize the casting and re-inspect it.

4.3.3.3.2.1.2 All body castings containing one or more magnetic particle indications shall be rejected. The Government reserves the right to have such castings examined by whatever means necessary to determine compliance or noncompliance with paragraph 3.5.2.

4.3.3.4 Hydrostatic Test (Alternative Body Assemblies).-Prior to gaging, each body assembly shall be subjected to an internal hydrostatic pressure to determine compliance with 3.7. Failure of any assembly to comply shall be cause for rejection of the assembly. Body assemblies which comply shall be stamped with the letter "H" as indicated on the assembly drawing.

4.3.3.5 Air pressure test (Alternative Body Assemblies).-Each body assembly that has passed the hydrostatic pressure test as indicated by the letter "H" stamped on the body, shall be subjected to a submerged air pressure test to determine compliance with 3.8. Failure of any assembly to comply as evidenced by the presence of air bubbles in the vicinity of the brazed joint or rising to the surface of the water shall be cause for rejection of the assembly. Body assemblies which comply shall immediately upon removal from the test fixture be stamped with the letter "A" as indicated on the assembly drawing.

4.3.3.6 Joint quality (Alternative Body Assemblies).-After air and hydrostatic testing but prior to the application of protective coating, 2 body assemblies from each lot shall be selected at random, sectioned and separated to determine compliance with paragraph 3.9. Failure of any specimen to comply shall be cause for rejection of the lot.

4.3.3.7 Protective coating.-The sample size and acceptance criteria shall be in accordance with the assembly drawing and the requirements of the specifications listed thereon. Failure to comply with the requirements of the applicable specifications shall be cause for taking the corrective action recommended therein.

4.3.4 Proving Ground functional tests.-The following Proving Ground tests will be performed by the Government.

4.3.4.1 First article sample.

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4.3.4.1.1 Initial production sample.-The quantities listed in Table I shall be ballistically tested in the phases indicated.

TABLE I

Phase	Quantity
Excess pressure	25
Service pressure	25

4.3.4.2 Production lot sample.-The quantities listed in Table II shall be ballistically tested in the phases indicated.

TABLE II

Phase	Quantity
Excess pressure	10 each lot
Service pressure	10 each lot

4.3.4.3 Excess pressure.-Samples provided in accordance with Tables I and II will be subjected to this test. Failure of any projectile assembly to comply with 3.11 shall be cause for rejection of the initial production sample or the production lot represented by the sample.

4.3.4.4 Service pressure.-Samples provided in accordance with Tables I and II will subjected to this test. Failure of any projectile assembly to comply with 3.11 shall be cause for rejection of the initial production sample or the production lot represented by the sample.

4.3.5 Inspection equipment.-Index of Inspection Equipment List Number IEL-10520203 identifies the inspection equipment required to perform the examination and tests prescribed herein. Equipment design shall be in accordance with the code designation specified on the applicable IEL.

4.4 Test methods and procedures.

4.4.1 Mechanical properties.

4.4.1.1 Hardness test.-Prior to finish machining, all bodies that will be subjected to the hardness test shall be prepared by grinding a flat on the cylindrical portion of the body where the obturator band groove will subsequently be machined. The test shall be performed using the equipment and procedures prescribed by Federal Test Method Standard No. 151, except that the minimum dwell time shall be 5 seconds when a Brinell Hardness Test is used.

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4.4.1.2 Tension test.-From the bodies selected in accordance with 4.3.3.2.2, two tension test specimens conforming to the largest obtainable standard "R" type specimens as prescribed by Federal Test Method Standard No. 151, shall be taken longitudinally from each side of the body, 180 degrees apart. Procedures and criteria for determining yield strength, elongation, reduction in area and tensile strength shall be as prescribed by Federal Test Method Standard No. 151 except that elongation shall be measured to the nearest thousandth prior to conversion into percent elongation. Percent elongation shall be recorded to the nearest tenth.

4.4.2 Metal defects (applicable to Body Alternative II and Body Alternative V).

4.4.2.1 Ultrasonic inspection.-The body castings shall be examined using ultrasonic equipment of the type and model approved by Frankford Arsenal, SMUFA-Q3000. Procedures for operation of the unit shall be as prescribed by Manual 10519685. Prior to start of production, the contractor shall submit a copy of his proposed procedure for complying with the requirements of the manual to Commanding Officer, Frankford Arsenal, Philadelphia, Pa. 19137, ATTN: SMUFA-Q3200, for written approval. After approval, no change in procedure shall be made without prior approval of the above organization.

4.4.2.2 Magnetic particle test procedure.-The procedures for performing this test shall be as specified in MIL-M-11472, for indirect magnetization by either the continuous method or the residual method. The solid content of the suspension shall be maintained at 0.2 to 0.4 milliliter (ml) of solids per 100 ml of suspension when oil is used, or 0.15 to 0.35 ml of solids per 100 ml of suspension when water is used. The suspension shall be checked at the start of each day's production and at intervals of not over 4 hours of continuous operation, utilizing the procedures and equipment prescribed by Section 4 of MIL-M-11472. When performing the test for solid content of the suspension, if it is determined that it does not comply, all body castings that have been tested and accepted since the last satisfactory test was performed shall be re-tested after the suspension has been brought to the proper level. The magnetic particle test shall be performed on equipment that complies with the requirements of PA-PD-792. The operator(s) of the equipment and inspector(s) making the determinations shall be certified in accordance with MIL-STD-410.

4.4.3 Hydrostatic Test (Alternative Body Assemblies).-The body assembly shall be hydrostatically tested using equipment capable of performing the test accurately and consistently. In performing the test, the pressure will be applied gradually until the required minimum pressure is reached, after which the pressure will be held for a minimum of 5 seconds. The equipment shall be calibrated prior to the start of each day's operation and at intervals of not over 2 hours of continuous operation. It shall also be calibrated prior to re-starting the operation after a layoff of one hour or more.

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4.4.4 Air pressure test (Alternative Body Assemblies).-This test shall be made using equipment capable of performing the test accurately and consistently. Body assemblies shall be mounted in an approved fixture in such a manner that the body assembly shall be completely submerged in water; the required air pressure shall be applied and held for a minimum of 15 seconds. The water in the tank shall be permitted to settle after immersion of the assemblies prior to the start of timing. Observation for evidence of leakage shall be made for the entire time the assembly is required to be pressurized. The equipment shall be calibrated at the start of each day's operation and at intervals of 4 hours and after a stoppage of 3 hours or more.

4.4.5 Joint quality (Alternative Body Assemblies).-The samples selected in accordance with paragraph 4.3.3.6 shall be sectioned and separated as follows: (a) the rear portion of the projectile assembly, 1-inch minimum forward of the base adapter, shall be removed in such a manner that no portion of the brazed joint will be damaged, (b) the rear portion of the body containing the base adapter shall then be cut vertically through the center of the base adapter, (c) each portion of the body shall be peeled from the base adapter section. After peeling, the brazed area of the two halves of the base adapter and the mating pieces of the body shall be visually examined to determine compliance with 3.9.

4.4.6 Protective coating.-The procedures for testing the cleaning of components and the application of protective coating shall be in accordance with the requirements of the applicable drawing and the specifications listed thereon. The salt spray test shall be performed utilizing equipment prescribed by Federal Test Method Standard No. 151.

4.4.7 Base cover weld or braze.-The test of the base cover weld or braze shall be performed as follows: Visually inspect the base shield and select the area that appears to have the weakest weld; insert a 5/8" chisel in this area and apply force by use of a 1-pound steel hammer at an approximate 1-foot stroke. If the weld breaks loose after 2 strokes, the body assembly will be classed defective. If the weld is found to be secure, the burr made by the chisel may be knocked back into position and accepted.

4.4.8 Proving Ground functional tests.

4.4.8.1 Excess pressure.-The samples supplied in accordance with Tables I and II inert loaded to the service weight and dummy fuzed will be assembled into complete rounds and fired for recovery in a mortar for which the round is standard. The propellant charge will be adjusted to produce 112 percent of the maximum rated pressure of the round in a new mortar.

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4.4.8.2 Service pressure.-The samples supplied in accordance with Tables I and II, inert loaded and fuzed will be assembled into complete rounds utilizing the service charge and fired in a mortar for which the round is standard for comparison with an equal number of rounds from a reference lot. The sample round and reference rounds will be fired alternately. Horizontal and vertical distances will be recorded for each round fired for determination of horizontal and vertical probable error (PE) in mils to assess accuracy. Data obtained will be recorded for information only and forwarded to the cognizant technical agency.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing and marking.-Packaging, Packing and Marking shall be in accordance with the requirements of level C in MIL-STD-1169.

5.2 Data cards.-Data card information shall be as specified in MIL-STD-1167. Data cards shall also denote weight group and range.

6. NOTES

6.1 Visual examination qualification.-When compliance with the applicable requirement is in doubt as a result of visual examination, this characteristic may be measured or gaged to determine acceptability.

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